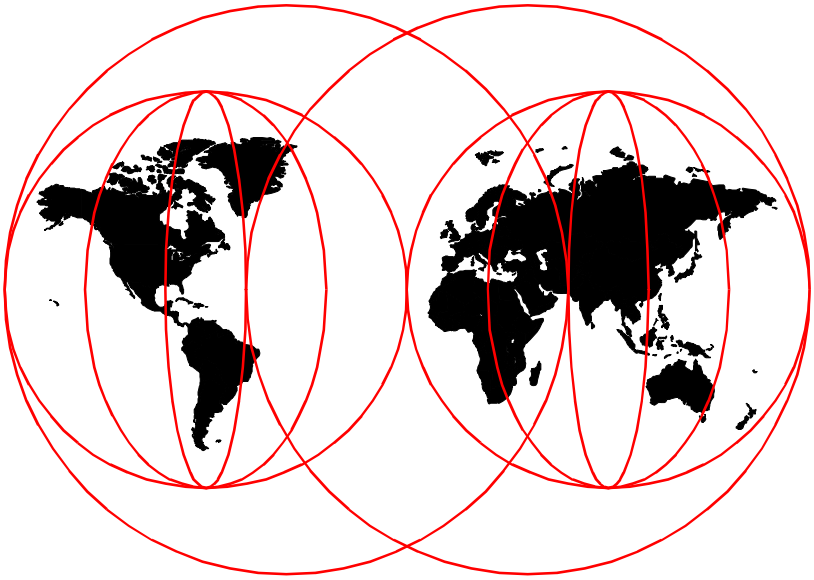
Getting Started with Data Warehouse and Business Intelligence

Maria Sueli Almeida, Missao Ishikawa, Joerg Reinschmidt, Torsten Roeber



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Getting Started with Data Warehouse and Business Intelligence

August 1999

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Before using this information and the product it supports, be sure to read the general information in Appendix B, “Special Notices” on page 225.

First Edition (August 1999)

This edition applies to Version 5, Release 1of DB2 Server for VSE & VM Program Number 5648-158, Version 5, Release 1of DB2 Server for OS/390 Program Number 5655-DB2, Version 5.2 of IBM Visual Warehouse, Version 1.0.1 of IBM DB2 OLAP Server, Version 2.1.2 of IBM Intelligent Miner for Data, Version 5.2 of IBM DB2 UDB for NT, Version 2.1.1of IBM DB2 DataJoiner Classic Connect, Version 2.1.3 of IBM DB2 Datajoiner for use with the Windows NT operating system, and Version 5.2 of IBM DB2 UDB for AIX, Version 2.1.2 of IBM Intelligent MIner for Data, Version 2.1.3 of IBM DB2 DataJoiner, Version 2.1.1 of IBM DB2 DataJoiner Classic Connect for use with the AIX Version 4.3.1 Operating System, CrossAccess VSE V4.1.

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Preface

This redbook describes an operating environment that supports Data Warehousing and BI solutions on all available platforms. The book describes how this environment was established. This project was carried out at the ITSO San Jose Center. DRDA over TCP/IP, DataJoiner Classic Connect, Cross Access for VSE, and Datajoiner are used to create this environment.

The objective of creating this environment was to have a base BI environment to support Data Warehouse, OLAP, and Data Mining solutions. This book is not intended to show how to set up and configure the different products used; some of these have been covered in previous product-specific redbooks.

This redbook is intended to be used in conjunction with the redbooks: *Intelligent Miner for Data: Enhance Your Business Intelligence*, SG24-5422 and *My Mother Thinks I'm a DBA! Cross-Platform, Multi-Vendor, Distributed Relational Data Replication with IBM DB2 DataPropagator and IBM DataJoiner Made Easy!,* SG24-5463. Future redbooks will enhance and exploit this environment.

The Team That Wrote This Redbook

This redbook was produced by a team of specialists from around the world working at the International Technical Support Organization San Jose Center.

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Thanks to the following people for their invaluable contributions to this project:

Paolo Bruni
International Technical Support Organization, San Jose Center

Scott Chen
International Technical Support Organization, San Jose Center

Thomas Groh
International Technical Support Organization, San Jose Center

Rick Long
International Technical Support Organization, San Jose Center

Wiliam Tomio Kanegae
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Thanks also to the professionals from Cross Access Corporation, for supporting and writing about the CrossAccess product.

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Chapter 1. What is BI ?

When reading about business intelligence (BI), one of the global definitions you may see is the one found on the IBM Business Intelligence Web page:

> "Business intelligence means using your data assets to make better business decisions. It is about access, analysis, and uncovering new opportunities."

Many of the concepts of business intelligence are not new, but have evolved and been refined based on experience gained from early host-based corporate information systems, and more recently, from data warehousing applications.

Given the increasing competition in today's tough business climate, it is vital that organizations provide cost-effective and rapid access to business information for a wide range of business users, if these organizations are to survive into the new millennium. The solution to this issue is a business intelligence system, which provides a set of technologies and products for supplying users with the information they need to answer business questions, and make tactical and strategic business decisions.

1.1 The Evolution of Business Information Systems

Inevitably the first question that arises when describing the objectives of a business intelligence system is, "Does a data warehouse have the same objectives and provide the same capabilities as a business intelligence system?" A similar question arose when data warehouses were first introduced, "Is a data warehouse similar to the corporate information systems and information centers we built in the past?" Although a quick and simple answer to both questions is "yes", closer examination shows that, just as there are important differences between a warehouse and early corporate information systems and information centers, there are also important differences between a business intelligence system and a data warehouse.

1.1.1 First-Generation: Host-Based Query and Reporting

Early business information systems employed batch applications to provide business users with the information they needed. The output from these applications typically involved huge volumes of paper that users had to wade through to get the answers they needed to business questions. The advent of terminal-driven time-sharing applications provided more rapid access to information, but these systems were still cumbersome to use, and required access to complex operational databases.

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This first generation of business information systems could, therefore, only be used by information providers, such as business analysts, who had an intimate knowledge of the data and extensive computer experience. Information consumers, like business executives and business managers, could rarely use these early systems, and instead had to rely on information providers to answer their questions and supply them with the information they needed.

1.1.2 Second-Generation: Data Warehousing

The second generation of business information systems came with data warehousing, which provided a giant leap forward in capability. Data warehouses have several advantages over first-generation systems:

- Data warehouses are designed to satisfy the needs of business users and not day-to-day operational applications.
- Data warehouse information is clean and consistent, and is stored in a form business users can understand.
- Unlike operational systems, which contain only detailed current data, the data warehouses can supply both historical and summarized information.
- The use of client/server computing provides data warehouse users with improved user interfaces and more powerful decision support tools.

1.1.3 Third-Generation: Business Intelligence

A data warehouse is still not a complete solution to the needs of business users. One weakness of many data warehouse solutions is that the vendors often focus on technology, rather than business solutions. While there is no doubt that data warehouse vendors provide powerful products for building and accessing a data warehouse, these products can require a significant amount of implementation effort.

The issue here is that warehouse products rarely come prepackaged for specific industries or application areas, or address particular business problems. This is very much like the situation in the early days of client/server computing, when vendors initially provided the technology for developing operational applications, but then quickly realized that organizations were looking for application and business solutions, and not yet more technology. Vendors fixed this problem, with the result that today many operational client/server applications are built using application packages, rather than being handcrafted by developers.

The same evolution has to happen in business information systems – vendors must provide application packages, and not just more technology. One distinguishing factor of business intelligence systems is that they focus on providing prepackaged application solutions in addition to improved technology.

Another issue with data warehousing is that much of the focus is still on building the data warehouse, rather than accessing it. Many organizations seem to think that if they build a data warehouse and provide users with the right tools, the job is done. In fact, it is just beginning. Unless the information in the warehouse is thoroughly documented and easy to access, complexity will limit warehouse usage to the same information providers as first-generation systems.

Business intelligence systems focus on improving the access and delivery of business information to both information providers and information consumers. They achieve this by providing advanced graphical- and Web-based online analytical processing (OLAP) and information mining tools, and prepackaged applications that exploit the power of those tools. These applications may need to process and analyze large volumes of information using a variety of different tools. A business intelligence system must, therefore, provide scalability and be able to support and integrate products from multiple vendors.

A business intelligence system may also simplify access to business information through the use an information catalog that documents decision support objects that can be employed by information consumers to answer the main business questions that arise in everyday business operations. Some also reduce the need for information consumers to access the warehouse at all. Instead, information consumers subscribe to the information they require, and the system delivers it to them at predefined intervals through a corporate intranet or e-mail.

The information stored in a data warehouse is typically sourced from operational databases (and in some cases external information providers). There is, however, also a considerable amount of business information kept in office and workgroup systems, on Web servers on corporate intranets and the public Internet, and in paper form on people's desks. To solve this issue, business intelligence systems are designed to support access to all forms of business information, not only the data stored in a data warehouse. Having a business intelligence system does not negate the need for a data warehouse – a data warehouse is simply one of the data sources that can be handled by a business intelligence system.

We see, then, that a business intelligence system is a third-generation business information system that has three key advantages:

1. Business intelligence systems not only support the latest information technologies, but also provide prepackaged application solutions.
2. Business intelligence systems focus on the access and delivery of business information to end users, and support both information providers and information consumers.
3. Business intelligence systems support access to all forms of business information, and not just the information stored in a data warehouse.

Figure 1 gives an overall view of the IBM business intelligence structure.

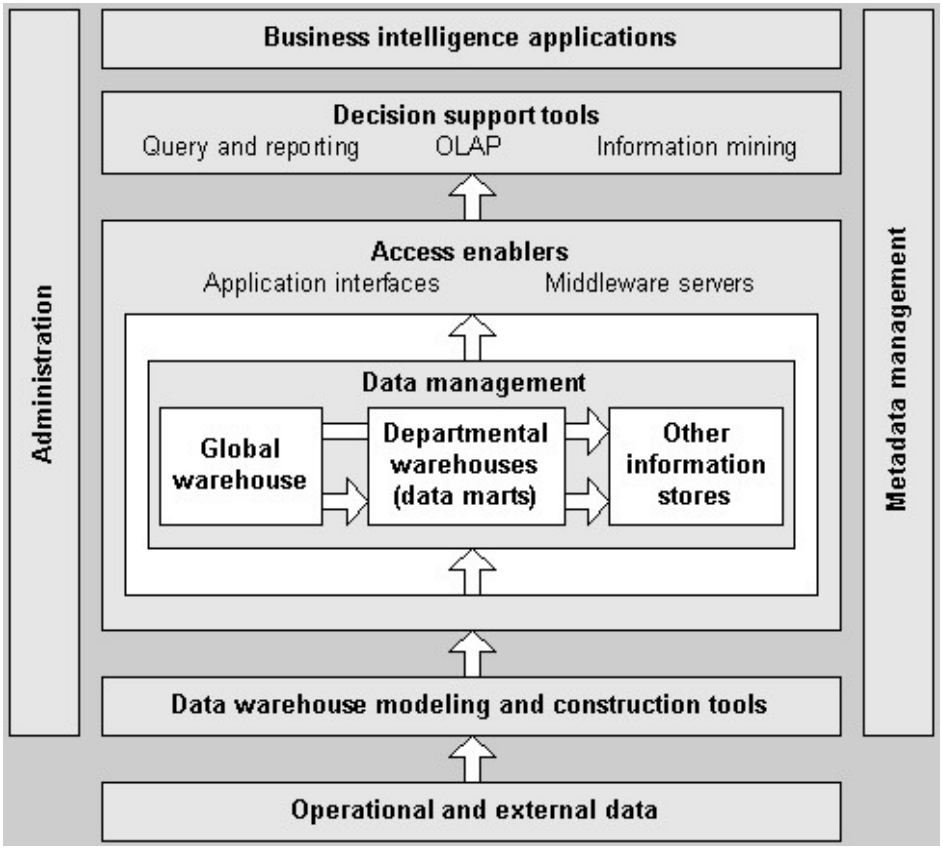

Figure 1. IBM Business Intelligence Structure

1.2 Why Do You Need Business Intelligence ?

Businesses collect large quantities of data in their day-to-day operations: data about orders, inventory, accounts payable, point-of-sale transactions, and of course, customers. In addition, businesses often acquire data, such as demographics and mailing lists, from outside sources. Being able to consolidate and analyze this data for better business decisions can often lead to a competitive advantage, and learning to uncover and leverage those advantages is what business intelligence is all about.

Some examples are:

- Achieving growth in sales, reduction in operating costs, and improved supply management and development.
- Using OLAP to reduce the burden on the IT staff, improve information access for business processing, uncover new sources of revenue, and improve allocation of costs.
- Using data mining to extract key purchase behaviors from customer survey data. NBA Coaches use IBM's Advance Scout to analyze game situations and gain a competitive advantage over other teams.

All of this is possible if you have the right applications and tools to analyze data, and more importantly, if the data is prepared in a format suitable for analysis. For a business person, it is most important to have applications and tools for data analysis, while for the IT community, having the tools to build and manage the environment for business intelligence is key.

1.3 What Do You Need for Business Intelligence ?

Business decision-makers have vastly different levels of expertise, and different needs for data analysis.

A wide range of approaches and tools are available, to meet their varied needs:

- Applications such as IBM's DecisionEdge for customer relationship management, and IBM's Business Discovery Series for data mining
- Query tools such as Impromptu and PowerPlay from Cognos, BusinessObjects from Business Objects, Approach from Lotus Development Corp., and IBM's Query Management Facility

- OLAP tools for multidimensional analysis such as Essbase from Arbor Software, and IBM's DB2 OLAP Server (developed in conjunction with Arbor)
- Statistical analysis tools such as the SAS System from SAS Institute Inc.
- Data Mining tools such as IBM's Intelligent Miner

Many of these applications and tools feature built-in support for Web browsers and Lotus Notes, enabling users to get the information they need from within a familiar desktop environment.

Most businesses do not want informal access to their operational computer systems, either for security or performance reasons. Rather, they prefer to build a separate system for business intelligence applications. To build these systems, tools are needed to extract, cleanse, and transform data from source systems which may be on a variety of hardware platforms, using a variety of databases. Once the data is prepared for business intelligence applications, the data is stored in a database system and must be refreshed and managed. These systems are called data warehouses or data marts, and the process of building and maintaining them is called data warehouse or data mart generation and management.

IBM's principal solution for generating and managing these systems is Visual Warehouse. Other IBM offerings such as the Data Replication Family and DataJoiner (for multi-vendor database access) can complement Visual Warehouse as data is moved from source to target systems. IBM also partners with companies such as Evolutionary Technologies International for more complex extract capabilities and Vality Technology Inc. for data cleansing technology.

For the warehouse database, IBM offers the industry leader: DB2. The DB2 family spans AS/400 systems, RISC System/6000 hardware, IBM mainframes, non-IBM machines from Hewlett-Packard and Sun Microsystems, and operating systems such as OS/2, Windows (95 & NT), AIX, HP-UX, SINIX, SCO OpenServer, and Solaris Operating Environment. When DataJoiner is used in conjunction with Visual Warehouse, non-IBM databases such as those from Oracle, Sybase, and Informix can be used as the warehouse database.

1.4 Business Driving Forces

So far we have seen that many of the driving forces behind business intelligence come from the need to improve ease-of-use and reduce the resources required to implement and use new information technologies. There are, however, three additional important business driving forces behind business intelligence:

1. The need to increase revenues, reduce costs, and compete more effectively. Gone are the days when end users could manage and plan business operations using monthly batch reports, and IT organizations had months to implement new applications. Today companies need to deploy informational applications rapidly, and provide business users with easy and fast access to business information that reflects the rapidly changing business environment. Business intelligence systems are focused towards end-user information access and delivery, and provide packaged business solutions in addition to supporting the sophisticated information technologies required for the processing of today's business information.
2. The need to manage and model the complexity of today's business environment. Corporate mergers and deregulation means that companies today are providing and supporting a wider range of products and services to a broader and more diverse audience than ever before. Understanding and managing such a complex business environment and maximizing business investment is becoming increasingly more difficult. Business intelligence systems provide more than just basic query and reporting mechanisms, they also offer sophisticated information analysis and information discovery tools that are designed to handle and process the complex business information associated with today's business environment.
3. The need to reduce IT costs and leverage existing corporate business information. The investment in IT systems today is usually a significant percentage of corporate expenses, and there is a need not only to reduce this overhead, but also to gain the maximum business benefits from the information managed by IT systems. New information technologies like corporate intranets and thin-client computing help reduce the cost of deploying business intelligence systems to a wider user audience, especially information consumers like executives and business managers. Business intelligence systems also broaden the scope of the information that can be processed to include not only operational and warehouse data, but also information managed by office systems and corporate Web servers.

1.5 Business Intelligence Requirements

Summarizing the two previous sections, we see that the main requirements of a business intelligence system are:

- Support for prepackaged application solutions.
- A cost-effective solution that provides a quick payback to the business and enables an organization to compete more effectively.
- Fast and easy access to an organization's business information for a wide range of end users, including both information providers and information consumers.
- Support for modern information technologies, including information analysis and discovery techniques like online analytical processing (OLAP) and information mining.
- An open and scalable operating environment.

Now that we have defined what a business intelligence system is, and have also identified its key requirements, we can move on to look at IBM's business intelligence strategy and products.

1.6 The IBM Business Intelligence Product Set

This part of the paper reviews the products and tools provided by IBM (and its key partners) for supporting a business intelligence software environment — these products are listed in Figure 2.

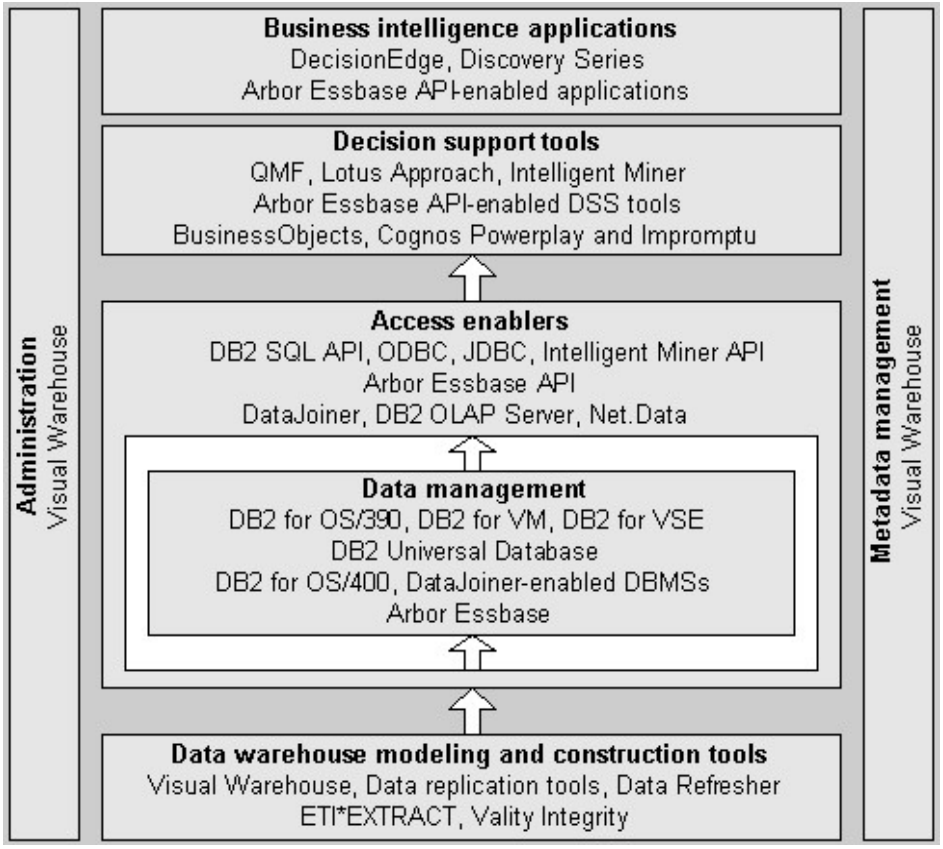

Figure 2. IBM Business Intelligence Product Set

1.6.1 Business Intelligence Applications

IBM's business intelligence applications are marketed under the DecisionEdge and Discovery Series brand names. DecisionEdge is a set of customer relationship management applications currently available for the telecommunications and utility industries. Each DecisionEdge offering provides integrated business applications, hardware, software, and consulting services. DecisionEdge for Telecommunications, for example, analyzes customer information (measuring profitability, predicting customer behavior, analyzing attrition, and so forth) and assists in the creation of tailored customer marketing programs. The product is used to design and deploy a customer information data warehouse on an IBM System/390, RS/6000 or AS/400 server using either an IBM DB2, Informix, Oracle or Sybase relational DBMS. Business users then employ DecisionEdge packaged and customized

applications to analyze customer information, and develop and monitor marketing programs.

Discovery Series is a set of applications and services that enable organizations to harness the power of IBM's Intelligent Data Miner product for customer relationship management. Included with the Discovery Series is the software, installation support, consulting, and training required to develop the data models, application-specific templates, and customized graphical output for an information mining application. The combination of services and prebuilt applications makes it easier for organizations to see quick results from information mining.

Business intelligence applications are also available for the DB2 OLAP Server (see description below). This product (which was developed by IBM and Arbor Software) employs the same API as Arbor Essbase, and can, therefore, be used with the many industry-specific third-party application packages available for Essbase.

1.6.2 Decision Support Tools

Business intelligence DSS tools can be broken down into three categories: query and reporting, online analytical processing (OLAP), and information mining.

1.6.2.1 Query and Reporting

The two main IBM query and reporting products are the Query Management Facility (QMF) and Lotus Approach. The System/390 version of QMF has been used for many years as a host-based query and reporting tool by DB2 for S/390 users (MVS, OS/390, VM, VSE). More recently, IBM introduced a native Windows version of QMF. Both versions support access not only to DB2 for S/390, but also any relational and non-relational data source supported by its DataJoiner middleware product (see description below). QMF host objects are compatible with QMF for Windows, extending the enterprise query environment to Windows and the Web. Output from QMF can be passed to other Windows applications like Lotus 1-2-3, Microsoft Excel, Lotus Approach, and many other desktop products via Windows OLE.

Lotus Approach is a desktop relational DBMS that has gained popularity due to its easy-to-use query and reporting capabilities. Approach can act as a client to a DB2 family database server, and, therefore, its query and reporting capabilities can be used to query and process DB2 data.

To increase the scope of its query and reporting offerings, IBM recently forged relationships with Business Objects (for its BusinessObjects product)

and Cognos (for its Impromptu and PowerPlay products). Like its key relationships with Evolutionary Technology International, Vality Technology, and Arbor Software, IBM intends the relationships with Business Objects and Cognos to be more than mere joint marketing deals — they also involve agreements to integrate the products from these companies with IBM's business intelligence offerings, for example, in the area of meta data interchange.

1.6.2.2 Online Analytical Processing (OLAP)

IBM's key product in the OLAP marketplace is the DB2 OLAP Server, which implements a three-tier client/server architecture for performing complex multidimensional data analysis. The middle tier of this architecture consists of an OLAP analytical server developed in conjunction with Arbor Software, which is responsible for handling interactive analytical processing and automatically generating an optimal relational star schema based on the dimensional design the user specifies. This analytical server runs on Windows NT, OS/2, or UNIX and can be used to analyze data managed by a DB2 Universal Database engine, or any relational database supported by DataJoiner. The DB2 OLAP Server supports the same client API and calculation engine as Arbor Essbase, and any of the many third-party GUI or Web-based tools that support Essbase can act as clients to the DB2 OLAP Server.

The value of the DB2 OLAP server lies in its ability to generate and manage relational tables that contain multidimensional data, in the available Essbase applications that support the product, and features within Visual Warehouse for automating the loading of the relational star schema with information from external data sources such as DB2, Oracle, Informix, IMS, and VSAM.

1.6.2.3 Information Mining

IBM has put significant research effort into its Intelligent Miner for Data product, which runs on Windows NT, OS/400, UNIX and OS/390, and can process data stored in DB2 family databases, any relational database supported by DataJoiner, and flat files. Intelligent Miner Version 1, released in 1996, enabled users to mine structured data stored in relational databases and flat files, and offered a wide range of different mining algorithms. Intelligent Miner Version 2 features a new graphical interface, additional mining algorithms, DB2 Universal Database exploitation, and improved parallel processing.

Intelligent Miner is one of the few products on the market to support an external API, allowing result data to be collected by other products for further analysis (by an OLAP tool, for example). Intelligent Miner has good data

visualization capabilities, and unlike many other mining tools, supports several information mining algorithms. IBM is also preparing to introduce its Intelligent Miner for Text product, which will provide the ability to extract, index, and analyze information from text sources such as documents, Web pages, survey forms, etc.

1.6.2.4 Access Enablers

Client access to warehouse and operational data from business intelligence tools requires a client database API. IBM and third-party business intelligence tools support the native DB2 SQL API (provided by IBM's Client Application Enablers) and/or industry APIs like ODBC, X/Open CLI, and the Arbor Essbase API.

Often, business information may be managed by more than one database server, and IBM's strategic product for providing access to this data is its DataJoiner middleware server, which allows one or more clients to transparently access data managed by multiple back-end database servers. This federated database server capability runs on Windows NT and UNIX, and can handle back-end servers running IBM or non-IBM data products, for example, IBM DB2 family, Informix, Microsoft SQL Server, Oracle, Sybase, VSAM, IMS, plus any ODBC, IBI EDA/SQL or Cross Access supported data source. Features of this product that are worthy of note include:

- Transparent and heterogeneous database access using a single dialect of SQL.
- Global optimization of distributed queries with query rewrite capability for poorly coded queries.
- Stored procedure feature that allows a global DataJoiner procedure to transparently access data or invoke a local procedure on any DataJoiner-supported database. This feature includes support for Java and Java Database Connectivity (JDBC).
- Heterogeneous data replication (using IBM DataPropagator, which is now integrated with DataJoiner) between DB2, Informix, Oracle, Sybase and Microsoft relational database products.
- Support for Web-based clients (using IBM's Net.Data product).

IBM's Net.Data Web server middleware tool (which is included with DB2) supports Web access to relational and flat file data on a variety of platforms, including the DB2 family, DataJoiner-enabled databases, and ODBC data sources. Net.Data tightly integrates with Web server interfaces, and supports client-side and server-side processing using applications written in Java, REXX, Perl, C++, or its own macro language.

1.6.2.5 Data Warehouse Modeling and Construction

IBM supports the design and construction of a data warehouse using its Visual Warehouse product family and data replication tools, and through third-party relationships with Evolutionary Technologies International (for its ETI*EXTRACT Tool Suite) and Vality Technology (for its Integrity Data Reengineering tool).

The Visual Warehouse product family is a set of integrated tools for building a data warehouse, and includes components for defining the relationships between the source data and warehouse information, transforming and cleansing acquired source data, automating the warehouse load process, and managing warehouse maintenance. Built on a DB2 core platform, Visual Warehouse can acquire source data from any of the DB2 family of database products, Informix, Microsoft, Oracle, Sybase, IMS databases, VSAM and flat files, and DataJoiner-supported sources.

Organizations have the choice of two Visual Warehouse packages, both of which are available with either Business Objects or Cognos add-ins for information access. The base package, Visual Warehouse, includes:

- DB2 Universal Database for meta data storage.
- A Visual Warehouse Manager for defining, scheduling, and monitoring source data acquisition and warehouse loading operations.
- A Visual Warehouse agent for performing the data capture, transformation and load tasks.
- The Visual Warehouse Information Catalog (formerly known as DataGuide) for exchanging meta data between administrators and business users.
- Lotus Approach for information access.

The second package, Visual Warehouse OLAP, adds the DB2 OLAP Server to the mix, allowing users to define and load a star schema relational database, as well as to perform automatic precalculation and aggregation of information as a part of the load process.

1.6.2.6 Data Management

Data management in the business intelligence environment is provided by IBM's DB2 family of relational DBMSs, which offers intelligent data partitioning and parallel query and utility processing on a range of multiprocessor hardware platforms. DB2 Universal Database on UNIX and Windows NT additionally supports both partition and pipeline parallelism, SQL CUBE and ROLLUP OLAP operations, integrated data replication,

dynamic bit-mapped indexing, user-defined types, and user-defined functions (which can be written in object-oriented programming languages such as Java).

1.7 The Challenge of Data Diversity

Rarely do medium- to large-sized organizations store all their enterprise and departmental data in a single database management system (DBMS) or file system. Instead, key business data is managed by multiple products (such as DB2, Oracle, Sybase, Informix, IMS, and others) running on different operating systems (MVS, UNIX, and Windows NT), often at widely separated locations. For business intelligence systems, the heterogeneity of most computing environments presents several challenges:

1. Providing users with an integrated view of the business is a key requirement for many business intelligence systems. Therefore, business intelligence software must be flexible enough to support diverse data sources.
2. Similarly, a robust business intelligence solution should support heterogeneous databases as target stores. Many companies are adopting a multi-tiered or distributed implementations, resulting in multiple data marts within a single company. With the increased adoption of distributed, client/server and departmental computing, it is common for different divisions of a company to use different database systems.
3. The foundation of an effective business intelligence solution is making it easy for the end user to access relevant business information. A user might need to consult the enterprise data warehouse for an answer to one question, but for a different question, query a local data mart — or for another, a correlation of data across a data mart and an external data source. Making the user aware of data location adds unneeded complexity, and discourages use of all available information.

IBM's business intelligence offerings meet the challenges of today's heterogeneous environments. For example, IBM's DataJoiner enables easy integration of diverse application, platform, and database environments. Introduced in 1995 and now in its second version, DataJoiner offers features to enable:

- **Transparent heterogeneous data access:** With DataJoiner, business users need not know that they are connecting to multiple databases. DataJoiner provides a single SQL interface that can read from and write to diverse data sources. DataJoiner manages the complexity of establishing connections to different data sources, translating requests into the native

interfaces of these data sources, coping with differences in data types, and determining an efficient data access strategy to satisfy any request. DataJoiner allows you to treat your data marts, external data, enterprise warehouse, and operational data store as a single, federated warehouse.

- **Simpler, more powerful queries:** DataJoiner makes queries in mixed and distributed data warehouse environments simpler and more powerful than ever. The ability to send a single query to access and join data stored in many IBM or non-IBM, relational or non-relational, local or remote data stores, combined with the power to employ global stored procedures, simplifies the creation of complex business intelligence applications. Further, DataJoiner features built-in support for Java and Java Database Connectivity (JDBC) for Java developers.
- **Optimized data access:** Fast response time to queries is important to all business users. But many of today's popular tools generate poorly structured queries, resulting in poor performance. DataJoiner includes sophisticated global query optimization processing, that rewrites queries automatically — transforming poorly structured queries into an efficient, logically equivalent form. The result? Queries perform better and are less costly to execute. From Web browsers or classic client, DataJoiner speeds results to demanding end users.
- **Heterogeneous replication:** Replication capability is vital in many data warehouse implementations. By enabling data warehouses to be continuously updated with changed and new operational data, business users are given access to the latest data. And, for companies that run their business systems 24x7, replication minimizes the reliance on batch windows for moving data from operational systems to data warehouses. IBM has integrated advanced replication capability into DataJoiner, enabling heterogeneous replication between DB2, Informix, Oracle, Sybase and Microsoft databases.